MILITARY SPECIFICATION

CONNECTORS, ELECTRICAL, MODULAR, AND COMPONENT PARTS GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for rack and panel connectors, base plate connectors, module connectors, cable connectors, and the component parts.

1.2 Classification.

1.2.1 Types. Connectors and assemblies covered by this specification shall be of the following type:

Type III	-	Aluminum base plate assembly with contact and bushing set.
Type IV	-	Module blade connector assembly on 100 mil grid.
Type V	-	Cable connector assembly, on 100 mil grid, with a dielectric block containing crimp or solder blade contacts.

1.2.2 Component parts. In addition this specification covers component parts utilized in type III through type V assemblies or in related applications.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-410	-	Plastic Polyamide (Nylon) Rigid, Rods, Tubes, Flats, Molded and Cast Parts.
QQ-A-200/8	-	Aluminum Alloy Bar, Rod, Shapes, Tube and Wire Extruded and Structural Shapes, 6061.
QQ-A-225/1	-	Aluminum Bar, Rod, and Wire; Rolled, Drawn, or Cold Finished, 1100.
QQ-A-225/6	-	Aluminum Alloy Bar, Rod and Wire, Rolled, Drawn, or Cold Finished, 2024.
QQ-A-225/8	-	Aluminum Alloy Bar, Rod, and Wire, and Special Shapes; Rolled, Drawn, or Cold Finished, 6061.
QQ-A-225/9	-	Aluminum Alloy Bar, Rod, Wire, and Special Shapes; Rolled, Drawn, or Cold Finish 7075.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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QQ-A-250/1 - Aluminum Alloy, 1100, Plate and Sheet.
QQ-A-250/2 - Aluminum Alloy, 3003, Plate and Sheet.
QQ-A-250/8 - Aluminum Alloy, 5052, Plate and Sheet.
QQ-A-250/9 - Aluminum Alloy, 5456, Plate and Sheet.
QQ-A-250/11 - Aluminum Alloy, 6061, Plate and Sheet.
QQ-A-591 - Aluminum Alloy Die Castings.
QQ-B-613 - Brass, Leaded and Nonleaded, Flat Products (Plate, Bar, Sheet and Strip).
QQ-B-626 - Brass, Leaded and Nonleaded, Rod, Shaved, Forgings, and Flat Products with Finished Edges (Bar and Strip).
QQ-B-750 - Bronze, Phosphor, Bar, Plate, Rod, Sheet, Strip, Flat Wire, and Structural and Special Shaped Sections.
QQ-C-530 - Copper-beryllium Alloy Bar, Rod, and Wire (Copper Alloy Numbers 172 and 173).
QQ-C-533 - Copper-beryllium Alloy Strip (Copper Alloy Numbers 170 and 172).
QQ-N-290 - Nickel Plating (Electrodeposited).
QQ-P-35 - Passivation Treatments for Austenitic, Ferritic, and Martensitic Corrosion-resisting Steel (Fastening Devices).
QQ-S-635 - Steel Plate, Carbon.
QQ-S-763 - Steel Bars, Shapes, and Forgings-corrosion Resisting.
QQ-S-765 - Steel Plate, Sheet, and Strip-corrosion Resisting.

MILITARY

MIL-M-14 - Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-C-5541 - Chemical Conversion Coatings on Aluminum and Aluminum Alloys.
MIL-H-7199 - Heat Treatment of Wrought Copper-beryllium Alloys, Process For (Copper Alloys Numbers 170, 172 and 175).
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-P-13949 - Plastic Sheet, Laminated, Metal Clad.
MIL-F-14072 - Finish For Ground Electronics Equipment.
MIL-W-16878 - Wire, Electrical, Insulated, High Temperature.
MIL-M-20693 - Plastic Molding Material, Polyamide (Nylon), Rigid.
MIL-I-43553 - Ink, Marking, Epoxy Base.
MIL-G-45204 - Gold Plating (Electrodeposited).
MIL-C-49055 - Cables, Special Purpose, Electrical (Flexible, Flat, Unshielded), (Round Conductor), General Specification For.
MIL-C-49059 - Cable, Electrical (Flexible, Flat, Unshielded), (Flat Conductor) General Specification For.
MIL-C-55330 - Connectors, Preparation For Delivery of.
MIL-W-81044 - Wire, Electric, Crosslinked Polyalkene, Crosslinked Alkane-Imide polymer or Polyarlene Insulated, Copper or Copper Alloy.

(See supplement 1 for the list of associated specification sheets).

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-202 - Plastic, Method of Testing.
FED-STD-406 - Plastics, Method of Testing.
FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1130 - Connections, Electrical, Solderless Wrapped.

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MIL-STD-1285	-	Marking of Electrical and Electronic Parts.
MIL-STD-1344	-	Test Methods, for Electrical Connectors.
MIL-STD-45662	-	Calibration Systems Requirements.
MS14033	-	Grid Patterns for Solderless Wrapped Electrical Connectors.
MS14034	-	General Contact Layout for Type III Base Plate Connector (.200 Grid).

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1-78	-	Surface Roughness, Waviness and Lay.
ANSI-Y14.5-1973	-	Dimensioning and Tolerancing for Engineering Drawings.
C7.9-1956(R1961)	-	Hard Drawn Copper Alloy Wires for Electrical Conductors, Specification for (ASTM B 105-55).
ANSI-H33.1-1973	-	Copper-Zinc-Lead (Leaded Red Brass or Hardware Bronze) Rod, Bar, and Shapes.

(Application for copies should be addressed to the American Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-A582-75	-	Free Machining Stainless and Heat Resisting Steel Bars, Hot-Rolled or Cold-Finished.
ASTM-A484-76	-	General Requirements for Stainless and Heat Resisting Wrought Steel Products (Except Wire).
ASTM-B140-72	-	Leaded Red Brass (Hardware Bronze) Rod, Bar and Shapes.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern (see 6.4).

3.2 Qualification. Connectors and accessories furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for the opening of the bids (see 4.5 and 6.5).

3.3 Material. The material for each part shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the connectors and accessories to meet the performance requirement. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3.1 Metals.

3.3.1.1 Metals. All metal parts (other than current carrying parts) exposed to environmental conditions shall be of a corrosion-resistant material or shall be suitably plated or chemically coated to resist corrosion during service life.

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3.3.1.2 Dissimilar metals. Where dissimilar metals are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided. Dissimilar metals shall be as defined in MIL-STD-889. Dissimilar metals such as brass, copper, or steel (except corrosion-resisting steel, passivated in accordance with 3.3.1.3) shall not be used in intimate contact with aluminum or aluminum alloys. However, spraying, coating, or plating of dissimilar base metals to provide similar or suitable abutting surfaces is permitted. See 6.7 for selection of compatable couples.

3.3.1.3 Corrosion resisting steel. Where applicable, corrosion resisting steel shall be 300 series, low magnetic permeability in accordance with QQ-S-635, QQ-S-763, or QQ-S-766, and passivated per QQ-P-35 or to finish E300 as specified in MIL-F-14072 (see 3.1).

3.3.2 Base plate (when applicable, see 3.1 and 3.4). The metal base plate for type III connectors shall be aluminum-alloy-sheet material in accordance with QQ-A-250/1, QQ-A-250/2, QQ-A-250/8, or QQ-A-250/11.

3.3.2.1 Base plate finish. The aluminum base plate shall be anodized in accordance with MIL-A-8625, or chemically treated in accordance with MIL-C-5541.

3.3.3 Insulation (dielectric).

3.3.3.1 Insulator bushings (used on type III). The insulator bushings for type III connectors shall be polyamide (nylon) in accordance with L-P-410 or type I of MIL-M-20693. The material shall be self-extinguishing in accordance with method 2021 of FED-STD-406.

3.3.3.2 Connector bodies (used on type IV and V). Connectors shall be glass-fiber-filled diallyl pthalate in accordance with type SDG-F of MIL-M-14 (see 3.1). The color shall be green or black.

3.3.4 Contacts.

3.3.4.1 Contact design and construction. Contacts shall be of the design, construction and physical dimensions specified (see 3.1).

3.3.4.2 Contact materials. Type III female contacts shall be made from (d.). Male blade type III contacts shall be made from a., b., or c. One piece wrappost contact and grounding bushing set for type III connectors shall be made from (e.). All other contacts (types IV and V) shall be one of the following materials (see 3.1):

a. Phosphor bronze alloy as specified in QQ-B-750, composition A, half hard.

b. Cadmium copper alloy, CABRA (Copper and Brass Association), No. 162, in accordance with the following requirements.

   1. 99.1 percent copper and 0.9 percent cadmium.

   2. conductivity, 80 percent.

   3. tensile strength of 55,000 to 65,000 pounds per square inch.

c. ANSI-H33.1 (ASTM-B-140), alloy B, half hard leaded red brass (hardware bronze), as specified in QQ-B-626 alloy 360 or QQ-B-613, composition 260.

d. Beryllium copper as specified in QQ-C-533, minimum Rockwell 45 on the 30N scale.

e. Beryllium copper as specified in QQ-C-530 minimum Rockwell 75 on 30T scale.

3.3.4.3 Contact plating (see 3.1). All contacts shall be gold plated in accordance with type II, grade C, class 1 of MIL-G-45204 to a thickness of 50 microinches minimum. Contacts shall have a minimum nickel underplating of 50 microinches in accordance with QQ-N-290, class 1. Localized finish specified by individual specification sheet.

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3.3.5 Housing and cover (used on type V).

3.3.5.1 Housing and cover. The housing and cover shall be either die cast aluminum in accordance with QQ-A-591, or extruded aluminum in accordance with QQ-A-200/8, or sheet aluminum in accordance with QQ-A-250/8 or QQ-A-250/9. The type aluminum used shall be suitable for the manufacturing technique employed.

3.3.5.2 Housing and cover finish. The housing and cover shall be anodized in accordance with MIL-A-8625. The basic metal shall be mechanically prefinished before anodizing to remove surface imperfections or roughness, and to assure uniformity of the finished surface. Minimum depth of anodic film shall be .0004 inch.

3.3.5.3 Housing and cover finish color (see 3.1). The housing and cover finish shall be black within the color range established by color chips 34052, 35042, 36076, 36081, 37038, and 37056 of FED-STD-595. Where red is specified, the color range shall be established by color chips 11302, 21302, 22190 and 12199 of FED-STD-595.

3.3.6 Screws. Holddown screws, clamp screws, and other screws used in the connector shall be of passivated, corrosion-resistant steel in accordance with QQ-S-763, class 302, 304, and 305, condition A.

3.3.7 Cable clamp (used on type V). Cable clamps shall be steel in accordance with QQ-S-635.

3.3.8 Keying pin (used on type IV and V). The keying pin shall be steel in accordance with ASTM-582 and ASTM-484A.

3.3.9 Grounding bushing two piece (used on type III). The grounding bushing shall be aluminum alloy in accordance with QQ-A-225/1, /6, /8, and /9.

3.3.9.1 Grounding bushing one piece (used on the type III). Grounding bushing for type III shall be beryllium copper in accordance with QQ-C-533.

3.3.9.2 Grounding bushing finish two piece. The aluminum alloy grounding bushing shall be chemically treated in accordance with MIL-C-5541.

3.3.10 Pin shields. Aluminum alloy 5052, H34, in accordance with QQ-A-250/8, anodized black in accordance with MIL-A-8625, type II or III, class 2 (see 3.1) alternate material. Unclad epoxy glass laminate in accordance with MIL-P-13949, type GF.

3.4 Design and construction. Type IV and V connectors and component parts of all type connectors covered by this specification shall be of the design, construction and physical dimensions specified (see 3.1 and 6.4). Dimensions and tolerances shall be in accordance with ANSI Y14.5-1973. Type III connectors are composed of the base plate, insulator bushing, contacts, and associated hardware (see 3.1). The grid pattern for the type III connectors shall be in accordance with MS14033. The design of the type III base plate connectors to accommodate type IV and V connectors shall conform to MS14034.

3.4.1 Threaded parts. Unless otherwise specified (see 3.1), all threaded parts shall be in accordance with FED-STD-H28

3.4.1.1 Engagement of threaded parts. All threaded parts shall engage by a length of at least 1-1/2 times the nominal diameter of the thread.

3.4.2 Connector assembly. No parts of the connector assembly shall be permanently displaced from their original, normal, fitted position (see 3.1) at completion of the specified tests.

3.4.3 Contacts. Contacts for type III and V connectors shall be designed as specified (see 3.1), and shall permit individual insertion and removal for the insulator or insulator bushing. The contact shall be designed to provide a means of holding or locking it in its applicable insulator or insulator bushing.

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3.4.3.1 Contact compliance. The contact shall be designed to assure proper operation without dependence on mechanical float, and to assure that minimum force (see 3.1) is transmitted to the connection joining the contact to the interconnecting media during mating and unmating. Contacts assembled to printed circuit boards shall not move or display looseness after assembly, or during and after mate and unmate cycles.

3.4.3.2 Wrappost (used on type III). Wrappost shall be defined in the individual specification sheets (see 3.1) and shall meet the wrappost requirements of MIL-STD-1130.

3.4.3.3 Crimp barrel. Crimp barrel shall be as defined in the individual specification sheet (see 3.1).

3.4.4 Inserts (used on type V). Type V connector inserts shall be interchangeable within the connector housing. To prevent improper assembly, inserts shall be polarized as specified (see 3.1).

3.4.4.1 Type IV connector inserts. Type IV connector inserts shall be fabricated as a solid dielectric.

3.4.4.2 Type V contact inserts. Type V contact insert assemblies shall be fabricated as either a solid single piece or a two piece solid dielectric. Both elements of a two piece insert assembly shall be polarized to insure that the assembly shall consist of one top and one bottom insert.

3.4.5 Base plate (when applicable, see 3.1 and 3.4). The base plate shall be as specified, and shall have punched or drilled holes into which insulator bushings and contacts can be inserted and retained.

3.4.6 Pin spacers. Pin spacers shall be specified (see 3.1).

3.5 Performance requirements.

3.5.1 Low level (dry circuit). When connectors are tested as specified in 4.7.2 the resistance shall not exceed 6.7 millohms (mΩ) per mated contact pair.

3.5.2 Electrical resistance.

3.5.2.1 Contact resistance. When connectors are tested as specified in 4.7.3.1, the voltage drop shall not exceed the specified value (see 3.1).

3.5.2.2 Crimp or solder joint resistance (type V). When crimp or solder terminated contacts are tested as specified in 4.7.3.2, the voltage drop between the wire and the contact pin shall not exceed 4 millivolts (mV).

3.5.2.3 Mounting plate to contact resistance (type III). When mounting plate to contact resistance is tested as specified in 4.7.3.3, the voltage drop between the mounting plate and contact shall not exceed 20 mV.

3.5.2.4 Insulation resistance. When connectors are tested as specified in 4.7.3.4, the insulation resistance shall be as specified in table I.

3.5.3 Engaging and separating forces (type III). When tested as specified in 4.7.4, the engaging force shall be 6 ounces maximum and the separating force shall be 2 ounces minimum.

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TABLE I. Insulation resistance.

Type	Resistance (megohms)					
	Contact to base plate			Contact to contact and contact to housing or card holder		
	Before humidity	Measured either in chamber or within 1 hour after removal from chamber	After 4 hours drying	Before humidity	Measured either in chamber or within 1 hour after removal from chamber	After 4 hours drying
III	10,000	10	500			
IV				100,000	100	18,000
V				50,000	50	10,000

3.5.4 Wrappost torque (type III). When tested as specified in 4.7.5, the contact shall withstand a torque of 3 ounce-inches for the .100 grid spacings without permanent rotation relative to the mounting plate or relative to the bushing.

3.5.5 Solderability (when specified, see 3.1). When contacts are tested as specified in 4.7.6, the dipped surface of the terminal shall be at least 95 percent covered with a continuous new solder coating.

3.5.6 Termination tensile strength (when applicable) (see 3.1). When terminated contacts are tensile tested as specified in 4.7.7, they shall meet the tensile strength specified in table II.

TABLE II. Tensile strength.

Wire size	Round wire per axial load (pounds) wire	Flat cable axial load pounds per inch of cable width
16	50	N/A
18	30	N/A
20	20	N/A
22	12	100
24	8	75
26	5	50
28	2.7	N/A
30	1.5	N/A
32	1.0	N/A

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3.5.7 Contact retention.

3.5.7.1 Type III connectors. When type III connectors are tested as specified in 4.7.8, no contact shall have been pushed out of its bushing, nor shall any bushing have been pushed out of the base plate. There shall be no visible movement of the contact or bushings.

3.5.7.2 Type IV and V connectors. When type IV and V connectors are tested as specified in 4.7.8, each contact as mounted in the connector shall withstand an axial force in either direction for a minimum of 10 seconds.

3.5.8 Temperature cycling. When connectors are tested as specified in 4.7.9, mated connector assemblies shall show no evidence of cracking, fracture or other damage detrimental to the operation of the connector.

3.5.9 Current overload. When contacts are tested as specified in 4.7.10, the contact plated surface shall not peel or discolor and the connectors shall meet the requirements of 3.5.2.1, 3.5.2.2, and 3.5.2.3.

3.5.10 Vibration. When mated connector assemblies are tested as specified in 4.7.11, there shall be no cracking or breaking, nor shall there be any loosening of parts, or other visible damage. There shall be no loss of electrical continuity of any of the contact circuits of more than 1 microsecond during the test. Connectors shall meet the requirements of 3.5.2, 3.5.3 and 3.5.7.

3.5.11 Shock (specified pulse). When mated connectors are tested as specified in 4.7.12, there shall be no breakage or loosening of contacts, cracking of inserts, nor other visible or dimensional damage which would cause electrical or mechanical breakdown. There shall be no loss of electrical continuity of any of the contact circuits of more than 1 microsecond during test.

3.5.12 Salt spray (corrosion). When connectors are tested as specified in 4.7.13, there shall be no visual signs of corrosion or corrosive salts on the basis metal. Connectors shall meet the requirements of 3.5.2 and 3.5.7.

3.5.13 Durability. When mated connectors are tested as specified in 4.7.14, there shall be no evidence of corrosion or corrosive salts that would adversely affect the connectors ability to function, and the voltage drop shall not exceed the specified value (see 3.1). In addition after completion of the test, type III connectors shall meet the requirements of 3.5.3 and 3.5.4.

3.5.14 Humidity. When tested as specified in 4.7.15, the connector shall meet the requirements of 3.5.2.4, 3.5.15, and 3.5.7.

3.5.15 Dielectric withstanding voltage. When unmated connector assemblies are tested as specified in 4.7.16, there shall be no evidence of arcing, breakdown, or damage.

3.5.16 Pin shield retention (type IV and V). When tested in accordance with 4.7.17, the pin shield shall withstand, without separation from the connector assembly or damage to the pin shield, a force of 4 pounds (17.8 Newtons) maintained for a minimum of 10 seconds.

3.5.17 Connector fin torque (type IV). When tested in accordance with 4.7.18, the connector assembly shall be capable of withstanding a 6 inch-pound torque applied in both directions along the fin in a direction perpendicular to the plane of the fin without detrimental effect on the mechanical or electrical structure of the assembly.

3.5.18 Fin cantilever load (type IV). When tested as specified in 4.7.19, the fin shall be capable of withstanding a force of 2 pounds applied perpendicular to the fin height.

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3.5.19 Guide rib strength test (type IV and V). When tested as specified in 4.7.20, the individual ribs shall withstand a torque of 10 inch-pounds maintained for a minimum of 10 seconds. There shall be no detrimental effect to the mechanical structure of the connector assembly.

3.5.20 Temperature-altitude. When tested as specified in 4.7.21, there shall be no evidence of cracks, burns, or other visible, or dimensional damage which would cause electrical or mechanical breakdown. In addition, at the completion of the temperature-altitude test, the connector shall meet the requirements of 3.5.15 and 3.5.2.4 (pre-humidity).

3.6 Assembly of contact and bushing sets (type III only). The contact and bushing sets when installed into the plate assembly shall satisfy the workmanship standards established by figures 1 and 3 herein, when visually inspected at a magnification of 5-10 power. Contacts shall be installed in the bushing in such a manner that contacts are flush or below the top surface as specified on figure 3. Bushing shall be free of damage on top surface, and configuration shall conform to figure 3. Figure 1A is a preferred condition. Figures 1B and 1C are unacceptable conditions. If more than 5 percent of the bushings display any conditions depicted in figures 1D through 1I, the plate is unacceptable.

3.7 Marking. Unless otherwise specified (see 3.1) connectors and component parts shall be permanently and legibly marked with the military part number, contact numbers (see 3.4.2), date code and manufacturer's source code in accordance with MIL-STD-1285. Other markings which in any way interfere with, obscure, or confuse those specified herein, are prohibited. All markings shall be made using epoxy base ink in accordance with MIL-I-43553 and shall remain legible after all tests. For components too small to mark, marking shall be on the packaging material (see 5.1). Use contrasting colors unless specified in the applicable specification sheet.

3.8 Workmanship. The workmanship on connectors supplied under this specification shall be suitable to assure conformance with the requirements contained herein. The contacts shall be free of burrs or sharp corners which would otherwise damage the plating when mating the connectors. Contacts and the connector body shall be free of cracks, voids, irregularities, etc. Housing cable clamps, holddown screws and inserts shall be free of burrs, sharp edges and flash.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality, and quantity to permit performance of the required inspection shall be established and maintained by the contractor. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

a. Materials inspection (see 4.4).

b. Qualification inspection (see 4.5).

c. Quality conformance inspection (see 4.6).

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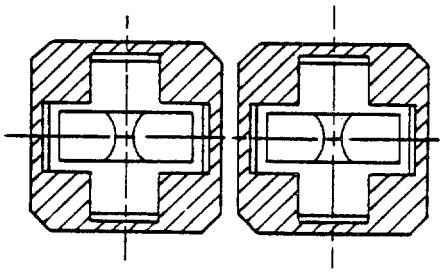


FIGURE 1A - Idealized condition.

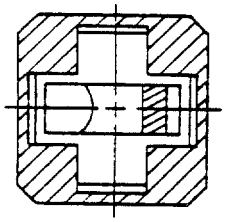


FIGURE 1B - Either tine broken.

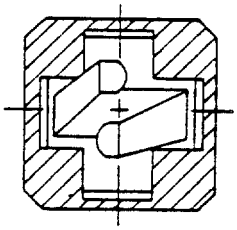


FIGURE 1C - Both contact tines twisted.

FIGURE 1. Contact and bushing set workmanship standards.

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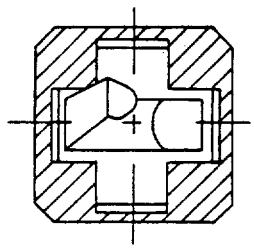


FIGURE 1D - Single contact tine twisted.

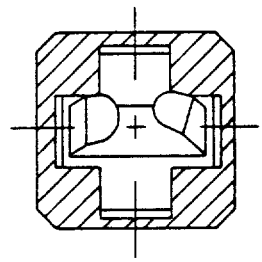


FIGURE 1E - Contact tines against bushing.

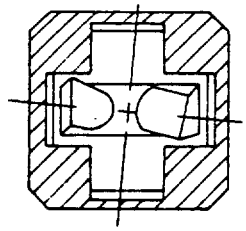


FIGURE 1F. Bushing rotated and contact tines twisted.

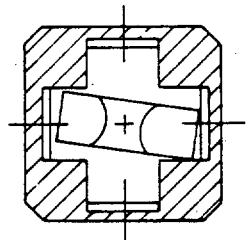


FIGURE 1G. Contact orientation not central to cruciform and contact tines bridging bushing.

FIGURE 1. Contact and bushing set workmanship standards - Continued.

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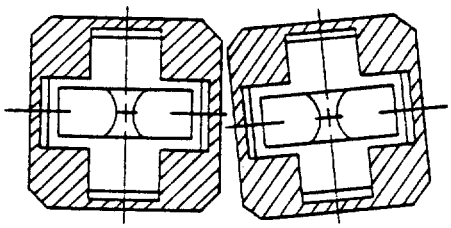


FIGURE 1H - Bushing and contacts with one set improperly oriented.

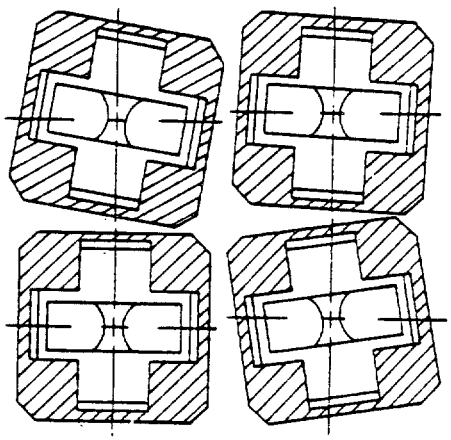


FIGURE 1I - Bushing and contacts randomly orientated in pattern.

FIGURE 1 - Contact and bushing workmanship standards - continued.

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4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the general requirements of MIL-STD-202, and MIL-STD-1344.

4.4 Materials inspection. Materials inspection shall consist of certification supported by verfying data that the materials listed in table III, used in fabricating the connectors, are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

TABLE III. Materials inspection.

Material	Requirement paragraph	Applicable specification
Aluminum	3.3.2, 3.3.2.1, 3.3.5.1, 3.3.5.2, 3.3.9, 3.3.9.1	QQ-A-225/1, QQ-A-225/6, QQ-A-225/8, QQ-A-250/1, QQ-A-250/2, QQ-A-250/8, QQ-A-250/9, QQ-A-250/11, MIL-A-8625, QQ-A-200/8, QQ-A-591
Copper alloy	3.3.4.2	QQ-C-533, QQ-B-750 QQ-B-626, QQ-B-513
Plastic	3.3.3.1, 3.3.3.2	L-P-410, MIL-M-20693, FED-STD-406, MIL-M-14
Nickel-plating	3.3.4.3	QQ-N-290
Gold-plating	3.3.4.3	MIL-G-45204
Steel	3.3.6, 3.3.7, 3.3.8	QQ-S-635, QQ-S-763
Marking ink	3.7	MIL-I-43553

4.5 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.5) on sample units produced with equipment and procedures normally used in production. Qualification inspection shall consist of the examinations and tests specified in table IV on qualification test samples specified in 4.5.1. After receipt of the letter of authorization from the agent responsible for qualification and after the applicant has conducted the examination and tests, the applicant shall submit three copies of his test reports (certified by the Government Inspection indicating the extent to which the tests were witnessed), the samples tested and a complete set of untested samples in accordance with 4.5.1.2 and 4.5.1.4 to the agent responsible for qualification.

4.5.1 Sample.

4.5.1.1 Sample size. Twelve connectors of types IV and V having the same number of contacts of the same size for which qualification is desired shall be subjected 40 qualification inspection. The sample shall be selected from a production run, and shall be produced with regular production equipment and procedures. The sample shall not be subjected to any screening test. The twelve connectors shall be divided equally between the various termination sizes, for which qualification is desired. 50 percent of each of the various termination sizes shall be wired to the smallest size wire and 50 percent of each of the various termination sizes shall be wired to the largest size conductor for which it is designed.

4.5.1.2 Type III sample. The qualification sample for type III shall be as indicated on test plates on figure 3. Three test plates of each type III configuration shall be tested. The figure 3 test plates shall contain samples of each of the (a) female contact and insulation bushing sets; (b) male contact and insulation bushing sets; (c) female contact and grounding bushing sets; (d) female contact and polarizing bushing sets; and (e) wrappost contact and grounding bushing sets for which qualification is desired. The wrappost length of the contact and bushing set installed in the test plates shall be 0.500 inch. Polarizing style S of the female contact and polarizing bushing sets shall be installed in the test plates.

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4.5.1.3 Preparation of sample (type V). The sample, which must be wired to complete the test, shall be wired as follows:

a. Crimp-solder-type contacts shall have the appropriate stranded wire terminated to the contact sizes indicated.

(1) Sizes 16-32--------Stranded tinned copper wire, minimum of 7 strands, in accordance with MIL-W-16878.

b. Solder-type contacts shall have the appropriate stranded wire or flat cable terminated to the contact sizes indicated.

(1) Sizes 22 and 24----Flat conductor cables, 20-conductor, .100 centers in accordance with MIL-C-49059.

(2) Sizes 24-30----Flat cable assembly, 20-conductor, in accordance with MIL-C-49055.

(3) Sizes 20-30-----Stranded copper wire, in accordance with MIL-W-81044 and MIL-W-16878.

4.5.1.4 Unassembled pieces. The following unassembled pieces shall be submitted to the agent responsible for qualification.

Type III

a. Ten of each wrappost lengths of each contact and bushing sets for which qualification is desired but not assembled on figure 3 test plates.

b. Ten of each polarizing styles of the female contact and polarized bushing sets for which qualification is desired but not assembled on figure 3 test plates. The wrappost length in the unassembled female contact and polarizing bushing sets may be any of the specified lengths (see 3.1).

c. Ten keying pegs of each configuration (figures 1A, 1B, and 1C of MIL-C-28754/39) of each material for which qualification is desired.

Type IV and V

a. Twenty contacts of each type IV and V connectors for which qualification is desired.

4.5.2 Inspection routine. Sample units shall be subjected to the qualification section specified in table IV, in the order shown. All sample units shall be subjected to the inspections of subgroup 1. The sample units shall then be divided as specified in table IV for the remaining subgroups and subjected to the inspection of their particular subgroup.

4.5.3 Failure. Failures in excess of those allowed in table IV shall result in refusal to grant qualification approval.

4.5.4 Retention of qualification. To retain qualification, the manufacturer shall forward to the qualifying activity at 12-month intervals, a summary of groups A and B. At 12-month intervals, a group C report shall be submitted. The qualifying activity shall establish the initial reporting date. The report shall consist of:

a. A summary of the results of the test performed for inspection of product for delivery, groups A and B, indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.

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b. The results of tests performed for periodic inspection, group C, including the number and mode of failures. The test report shall include results of all periodic inspection test performed and completed during the 12-month period. If the test results indicate nonconformance with specification requirements and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 60 days after the end of each 12-month period may result in loss of qualification for the product. In addition to the periodic submission of inspection data, the contractor shall immediately notify the qualifying activity at any time during the 12-month period that the inspection data indicated failure of the qualified product to meet the requirements of this specification (see 4.6.2.1).

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods that has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit a representative product of each part number to testing in accordance with the qualification inspection requirements.

TABLE IV. Qualification inspection.

Inspection	Connector type			Requirement paragraph	Method paragraph	Number of units to be inspected	Number of defectives allowed
	III	IV	V				
Subgroup I							
Visual and mechanical examination	X	X	X	3.1, 3.3, 3.4, 3.6, 3.7, 3.8	4.7.1		
Dielectric withstanding voltage	X	X	X	3.5.15	4.7.16		
Insulation resistance	X	X	X	3.5.2.4	4.7.3.4		
Low level (dry circuit)	X	X	X	3.5.1	4.7.2		
Contact resistance	X	X	X	3.5.2.1	4.7.3.1		
Crimp or solder joint resistance			X	3.5.2.2	4.7.3.2	12/3*	0
Mounting plate to contact resistance	X			3.5.2.3	4.7.3.3		
Engaging and separating forces	X			3.5.3	4.7.4		
Wrappost torque	X			3.5.4	4.7.5		
Contact retention	X	X	X	3.5.7	4.7.8		
Solderability	X	X	X	3.5.5	4.7.6		

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TABLE IV. Qualification inspection - Continued.

Inspection	Connector type			Requirement paragraph	Method paragraph	Number of units to be inspected	Number of defectives allowed
	III	IV	V				
Subgroup II							
Temperature cycling -	X	X	X	3.5.8	4.7.9		
Current overload- - -	X	X	X	3.5.9	4.7.10		
Contact resistance -	X	X	X	3.5.2.1	4.7.3.1		
Mounting plate to contact resistance -	X			3.5.2.3	4.7.3.3		
Crimp or solder joint resistance- - - - - -			X	3.5.2.2	4.7.3.2		
Vibration - - - - - -	X	X	X	3.5.10	4.7.11	4/1*	0
Contact retention- -	X	X	X	3.5.7	4.7.8		
Contact resistance -	X	X	X	3.5.2.1	4.7.3.1		
Mounting plate to contact resistance -	X			3.5.2.3	4.7.3.3		
Engaging and separating forces- - - - -	X			3.5.3	4.7.4		
Pin shield retention-		X	X	3.5.16	4.7.17		
Visual and mechanical examination - - - - -	X	X	X	3.1, 3.3, 3.4, 3.6, 3.7, 3.8	4.7.1		
Subgroup III							
Durability- - - - - -	X	X	X	3.5.13	4.7.14		
Engaging and separating forces- - - - -	X			3.5.3	4.7.4		
Wrappost torque - - -	X			3.5.4	4.7.5		
Contact retention - -	X	X	X	3.5.7	4.7.8		
Salt spray (corrosion) - - - - - - - -	X	X	X	3.5.12	4.7.13	4/1*	0
Contact resistance- -	X	X	X	3.5.2.1	4.7.3.1		
Mounting plate to contact resistance- -	X			3.5.2.3	4.7.3.3		
Visual and mechanical examination - - - - -	X	X	X	3.1, 3.3, 3.4, 3.6, 3.7, 3.8	4.7.1		

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TABLE IV. Qualification inspection - Continued.

Inspection	Connector type			Requirement paragraph	Method paragraph	Number of units to be inspected	Number of defectives allowed
	III	IV	V				
Subgroup IV							
Shock (specified pulse)	X	X	X	3.5.11	4.7.12		
Insulation resistance	X	X	X	3.5.2.4	4.7.3.4		
Humidity	X	X	X	3.5.14	4.7.15		
Insulation resistance	X	X	X	3.5.2.4	4.7.3.4		
Dielectric withstanding voltage	X	X	X	3.5.15	4.7.16		
Contact retention	X	X	X	3.5.7	4.7.8		
Pin shield retention		X	X	3.5.16	4.7.17	4/1*	0
Connector fin torque		X	X	3.5.17	4.7.18		
Fin cantilever load		X		3.5.18	4.7.19		
Guide rib strength test		X	X	3.5.19	4.7.20		
Visual and mechanical examination	X	X	X	3.1, 3.3, 3.4, 3.6, 3.7, 3.8	4.7.1		
Temperature altitude		X	X	3.5.20	4.7.21		
Dielectric withstanding voltage		X	X	3.5.15	4.7.16		
Insulation resistance (pre-humidity)		X	X	3.5.2.4	4.7.3.4		
Insulation resistance	X	X	X	3.5.2.4	4.7.3.4		
Subgroup V							
Visual and mechanical examination	X	X	X	3.1, 3.3, 3.4, 3.6, 3.7, 3.8	4.7.1	20 contacts	0
Termination tensile strength			X	3.5.6	4.7.7		

* Indicates inspection quantity for type III connectors only.

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4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of groups A and B inspection. Delivery of products which have passed the groups A and B inspection shall not be delayed pending the results of the qualification inspection.

4.6.1.1 Inspection lot. An inspection lot shall consist of all connectors having the same number and size of contacts, of the same type and contact spacing, or their component parts, whichever is being procured, produced under essentially the same conditions and offered for inspection at one time.

4.6.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table V.

4.6.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in table V.

TABLE V. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective)
Visual and mechanical - -	3.1, 3.3, 3.4, 3.6, 3.7, 3.8	4.7.1	2.5

4.6.1.2.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.6.1.3 Group B inspection. Group B inspection shall consist of the inspections specified in table VI, in the order shown, and shall be made on sample units which have been subjected to and have passed the group A inspection.

4.6.1.3.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for special inspection level S-4. The acceptable quality levels (AQL) shall be as specified in table VI. Major and minor defects shall be as defined in MIL-STD-105.

TABLE VI. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective)
Contact resistance - - - - -	3.5.2.1	4.7.3.1	0.65
Mounting plate to contact resistance (type III) - - -	3.5.2.3	4.7.3.3	
Engaging and separating forces (type III) - - - - -	3.5.3	4.7.4	
Insulation resistance- - - -	3.5.2.4	4.7.3.4	
Contact retention- - - - - -	3.5.7	4.7.8	

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4.6.1.3.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall clearly be identified as reinspected lots.

4.6.1.3.3 Disposition of sample units. Sample units which have been subjected to group B inspection may be delivered on the contract or purchase order, if the lot is accepted.

4.6.2 Periodic inspection. Periodic inspection shall consist of group C inspection. Where the results of these inspections show noncompliance with the applicable requirements (see 4.6.2.1.4), the delivery of products that have passed groups A and B shall not be delayed pending the results of these periodic inspections.

4.6.2.1 Group C inspection. Group C inspection shall consist of the inspections specified in table IV, in the order shown. Group C inspection shall be made on sample units which have passed the groups A and B inspections.

4.6.2.1.1 Sampling plan. The group C samples shall be in accordance with 4.5.1.

4.6.2.1.2 Failures. If one or more sample units fail to pass group C inspection, the sample shall be considered to have failed.

4.6.2.1.3 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract.

4.6.2.1.4 Noncompliance. If a sample fails to pass group C inspection, the manufacturer shall notify the qualifying activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the qualifying activity). Groups A and B inspection may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

4.7 Methods of examination and test.

4.7.1 Visual and mechanical examination. Connectors shall be examined to determine that the material, design, construction, physical dimensions, marking and workmanship are in accordance with the applicable requirements (see 3.1, 3.3, 3.4, 3.6, 3.7, and 3.8).

4.7.2 Low level (dry circuit) (see 3.5.1). Connectors shall be tested in accordance with method 3002 of MIL-STD-1344.

4.7.3 Electrical resistance.

4.7.3.1 Contact resistance (see 3.5.2.1). Connectors shall be wired and tested in accordance with method 3004 of MIL-STD-1344. The following details and exceptions shall apply:

a. Method of connection, see figure 2.

b. Test current, 3 amperes, unless otherwise specified (see 3.1).

c. Number of activations prior to measurement - 1.

d. Number of test activations - 1.

e. Number of measurements per activation - 1.

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4.7.3.2 Crimp or solder joint resistance (see 3.5.2.2). Connectors with crimp or solder terminated contacts (see 3.1) shall be wired as shown on figure 4 and tested in accordance with method 3004 of MIL-STD-1344. The following details and exceptions shall apply:

a. Method of connection, see figure 4.

b. Test current, see 3.1.

c. Number of measurements per activation - 1.

4.7.3.3 Mounting plate to contact resistance (type III) (see 3.5.2.3). The test sample shall conform to the requirements of 3.5.2.3 when tested as shown on figure 5. The test current shall be 3 amperes.

4.7.3.4 Insulation resistance (see 3.5.2.4). Connectors shall be tested in accordance with method 3003 of MIL-STD-1344. The following details shall apply:

a. Test voltage: 100 V dc ±5 percent.

b. Points of measurement:

Type III - Contact to base plate.

Types IV and V - Contact to contact and contact to housing or card holder.

4.7.4 Engaging and separating forces (type III) (see 3.5.3). With the test sample secured to the bed of a testing machine, determine the maximum engaging and minimum separating forces for contacts. The engaging and separating forces shall be determined using method 2014 of MIL-STD-1344, with the test blade shown on figure 6. Each contact shall meet the requirements of 3.5.3.

4.7.5 Wrappost torque (type III) (see 3.5.4). The assembled contact and bushing shall be tested in accordance with MIL-STD-202, method 211, test condition E, and shall meet the requirement of 3.5.4.

4.7.6 Solderability (type IV and V) (when specified, see 3.1 and 3.5.5). Terminals shall be tested in accordance with method 208 of MIL-STD-202.

4.7.7 Termination tensile strength (type V) (see 3.5.6). Contacts shall be tested in accordance with method 2003 of MIL-STD-1344.

4.7.7.1 Termination tensile strength (round wire, type V connector only) (see 3.5.6). The axial load specified in table II shall be applied to each individual wire. Samples of the terminated contact-wire shall be placed in a tensile testing device and sufficient force applied to pull the wire out of the contact or to break the wire. The terminated joint of the contact shall not break or become distorted before the minimum tensile strength specified is reached. The speed of head travel of the tensile device shall be 1 ±.250 inch per minute.

4.7.7.2 Termination tensile strength (flat conductor cable only, type V connector) (see 3.5.6). Unmated connectors shall be tested in accordance with method 2003 of MIL-STD-1344. The following details and exceptions shall apply:

a. Axial force shall be in accordance with table II and shall be applied uniformly across the width of the cable.

b. The cable shall be clamped 6 (six) inches from the mating face of the assembly.

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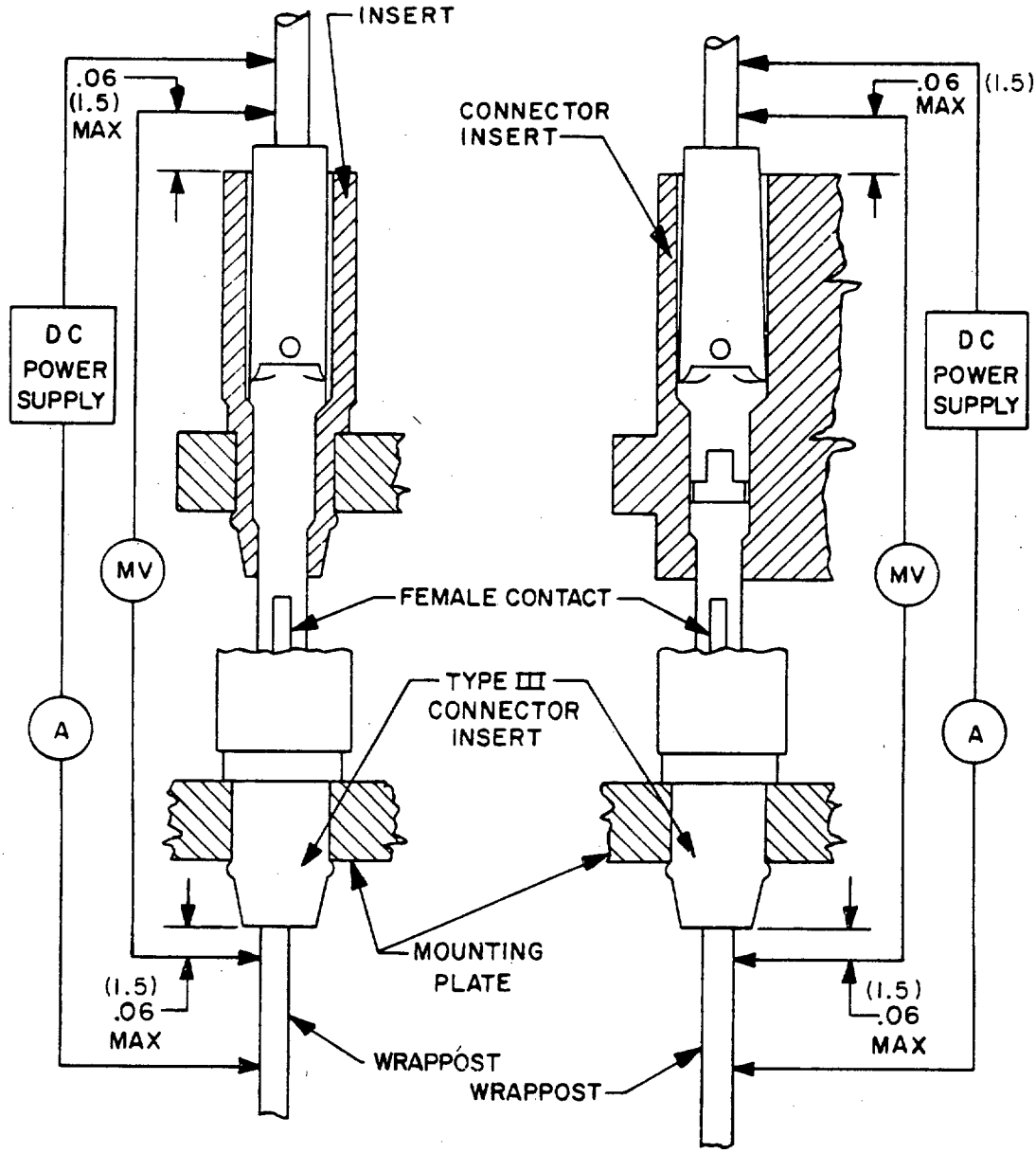


FIGURE 2. Test setup for measuring mated contact resistance.

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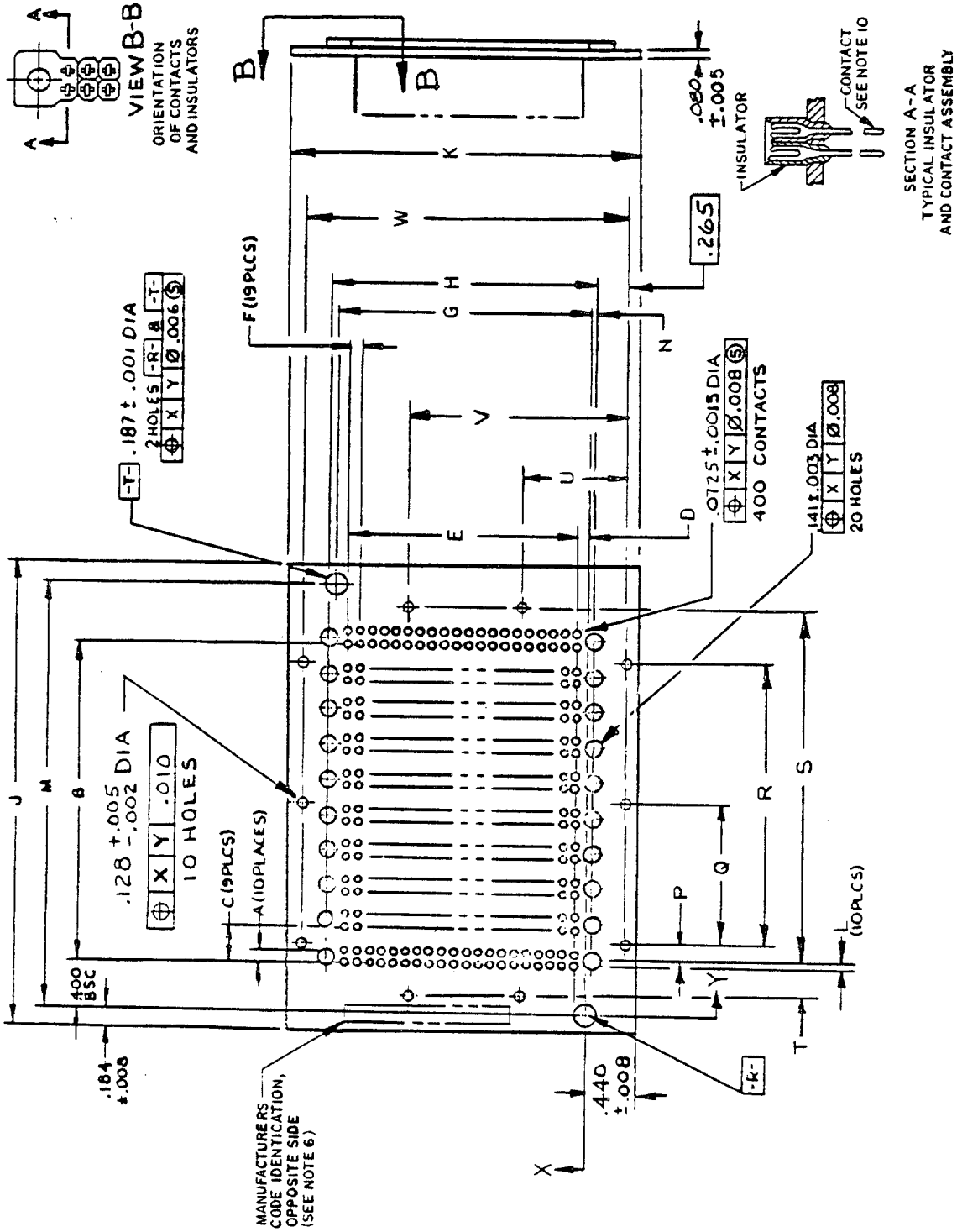


FIGURE 3. Base plate grid with .100 spacing.

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GRID SPACING	A BSC	B BSC	C BSC	D BSC	E BSC	F BSC	G BSC	H BSC	J ±.01	K ±.01	L BSC	M BSC	N BSC	P BSC	Q BSC	R BSC	S BSC	T BSC	U BSC	V BSC	W BSC
.100	.100	2.700	.300	.100	1.900	.100	2.100	2.200	3.97	2.98	.050	3.600	.050	.150	1.200	2.400	3.000	.300	.865	1.865	2.730

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.001	.02	.075	1.90	.265	6.73	1.800	45.72	2.730	69.34	3.750	95.25
.0015	.04	.080	2.03	.300	7.62	1.865	47.37	2.750	69.85	3.830	97.28
.002	.05	.100	2.54	.375	9.52	1.900	48.26	2.850	72.39	3.970	100.83
.003	.08	.125	3.18	.400	10.16	2.100	53.34	2.98	75.69	4.050	102.87
.005	.13	.128	3.25	.440	11.17	2.187	55.55	3.000	76.20	4.080	103.63
.006	.15	.141	3.58	.450	11.43	2.200	55.88	3.150	80.01	4.300	109.22
.008	.20	.150	3.81	.865	21.97	2.375	60.32	3.280	83.31	4.500	114.30
.010	.25	.184	4.67	1.093	27.76	2.400	60.96	3.300	83.82	4.670	118.61
.050	1.27	.187	4.74	1.200	30.48	2.553	64.85	3.375	85.72	5.000	127.00
.0625	1.59	.188	4.77	1.277	32.44	2.625	66.68	3.530	89.66	5.370	136.39
.0725	1.84	.225	5.71	1.500	38.10	2.700	68.58	3.600	91.44		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Coating, anodic MIL-A-8625, type II, black seal or chemical film per MIL-C-5541, class 3, aluminum plate only.
4. All holes shall meet the dimensional requirements specified on the drawing for a minimum of 50% of the plate thickness. Delineation and dimensions of remaining portion of holes optional as long as minimum opening is maintained.
5. Plate shall be flat within .003 per inch.
6. Plate shall be marked as per drawing using MIL-I-43553.
   a. The part number and the manufacture's code identification is in area outlined by phantom lines using .12 ± 02 high Gothic or Futura characters.
   b. Lines .06 ± 02 wide as shown.
7. The .187 diameter holes -R- and -T- shall be used to locate the plate in the wire wrapping machine.
8. Dimensions are taken from datums X and Y unless otherwise specified.
9. Assembly of contacts, insulators and bushings shall meet the requirements of MIL-C-28754 and specified slash sheet.
10. The centerline at the tip of the wrappost shall be | ⌖ | X | Y | 0.020 |.

FIGURE 3. Base plate grid with .100 spacing-continued.

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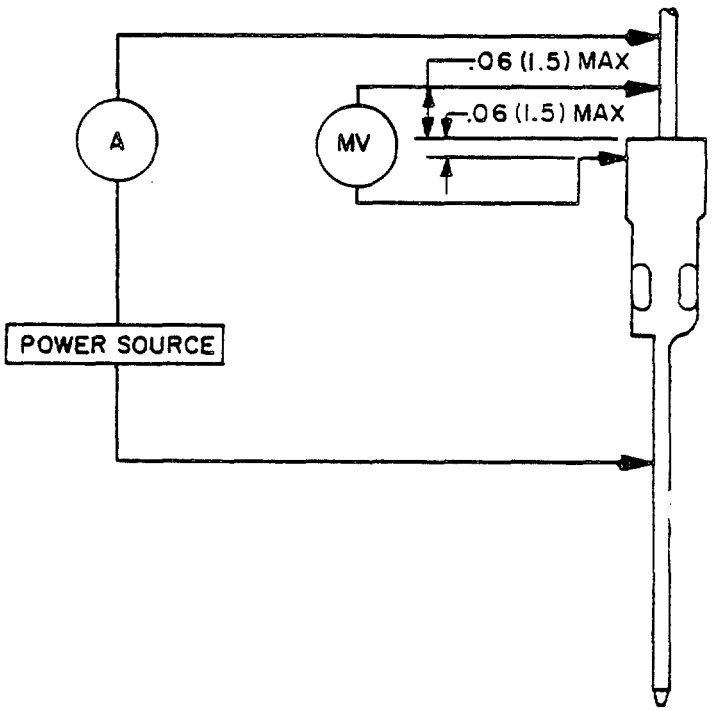


FIGURE 4. Test setup for measuring crimp or solder joint resistance.

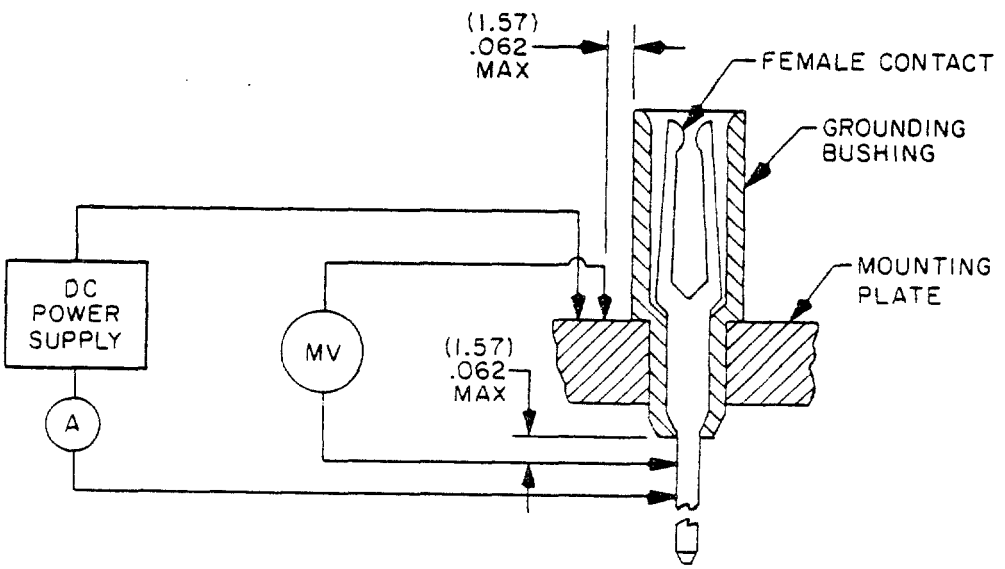
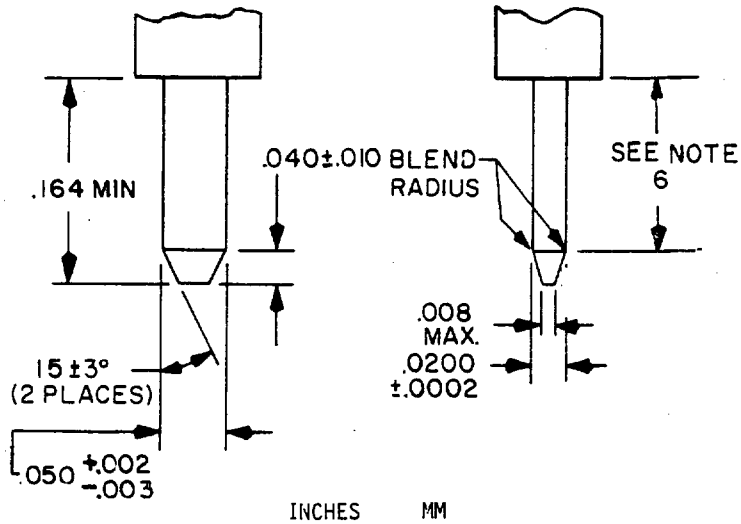


FIGURE 5. Test setup for measuring mounting plate to contact resistance.

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INCHES	MM
.002	.05
.003	.08
.005	.13
.008	.20
.010	.25
.020	.51
.040	1.02
.050	1.27
.164	4.17
.200	5.08

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Material: Hardened tool steel.
4. Hardness: Rockwell "C" 50-55.
5. Surface finish: 4-8 micro-inch on working surfaces in accordance with MIL-STD-10.
6. Grinding to be in the direction of blade movement during test.

FIGURE 6. Contact engaging and separating, test blade.

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4.7.8 Contact retention (see 3.5.7). Connectors shall be tested in accordance with method 2007 of MIL-STD-1344. The following details and exceptions shall apply:

a. Number of samples - A minimum of 10 contacts per test specimen shall be tested.

b. Applied axial load - One pound per second until a 7-1/2 pound load has been reached. Maintain load for 10 seconds. (When hook-up wire is used, the wire shall be large enough to withstand the applied load.)

c. Maximum allowable contact displacement during application of specified force and after removal of specified force - None.

d. Axial direction - Straight.

4.7.9 Temperature cycling (see 3.5.8). Connectors, wired (when applicable) and mated, shall be tested in accordance with method 1003 of MIL-STD-1344. The following details and exceptions shall apply:

a. Test condition letter - A, except maximum temperature extreme shall be 105°C.

b. Measurements before and after cycling - Not applicable.

Following this test, connectors shall meet the requirements of 3.5.8.

4.7.10 Current overload (see 3.5.9).

4.7.10.1 Mated contacts. Mated contacts shall be subjected to current overloads of 125 percent of rated current (see 3.1) for 2 hours, and 150 percent of rated current of contacts for 5 minutes. After being cooled to room temperature, connectors shall be tested in accordance with 4.7.3.1.

4.7.10.2 Crimped or solder joint. The wired crimped or solder joint shall be subjected to current overloads of 125 percent of rated current (see 3.1) for 2 hours, and 150 percent of rated current of contacts for 5 minutes. After being cooled to room temperature, connectors shall be tested in accordance with 4.7.3.2.

4.7.10.3 Mounting plate to contact. The mounting plate to contact joint shall be subjected to current overloads of 125 percent of rated current (see 3.1) for 2 hours, and 150 percent of rated current of contacts for 5 minutes. After being cooled to room temperature, connectors shall be tested in accordance with 4.7.3.3.

4.7.11 Vibration (see 3.5.10). Mated connector assemblies shall be tested in accordance with method 2005 of MIL-STD-1344. The following details and exceptions shall apply:

a. Test condition III and VE.

b. Duration of the random vibration test (VE) shall be 15 minutes per axis.

Following this test, connectors shall meet the requirements of 3.5.10.

4.7.12 Shock (specified pulse) (see 3.5.11). Connectors shall be tested in accordance with method 2004 of MIL-STD-1344. The following details and exceptions shall apply:

a. Test condition G.

b. Measurements during and after test - The mated connector assemblies shall be monitored to determine electrical discontinuity by a method which shall at least be sensitive enough to monitor or register, automatically, any electrical discontinuity of 1.0 microsecond.

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4.7.13 Salt spray (corrosion) (see 3.5.12). Connectors shall be tested in accordance with method 1001 of MIL-STD-1344. The following details and exceptions shall apply:

a. Applicable salt solution - 5 percent.

b. Test condition letter - B.

c. Conditioning after exposure - All exposed surfaces of connectors shall be thoroughly washed and dried in a circulating air oven at a temperature of 38°C ±3°C.

d. Measurements after exposure - Connectors shall be tested in accordance with 4.7.3 and 4.7.8. Connectors shall meet the requirements of 3.5.12.

4.7.14 Durability (see 3.5.13). Connector assemblies complete with wire, contacts, applicable hoods, clamp and associated hardware shall be subjected to 500 cycles of mating and unmating in accordance with method 2016 of MIL-STD-1344. Testing of connectors shall not exceed 500 cycles per hour. Connectors shall be tested in accordance with 4.7.4 and 4.7.5. Following this test, mated connectors shall meet the requirements of 3.5.13.

4.7.15 Humidity (see 3.5.14). The test sample shall be humidity tested in accordance with MIL-STD-1344, method 1002, type III. Following the test, connectors shall be tested in accordance with 4.7.3.4, 4.7.16, 4.7.8, and shall meet the requirements of 3.5.14.

4.7.16 Dielectric withstanding voltage (see 3.5.15). Connector assemblies shall be tested in accordance with method 3001 of MIL-STD-1344. The following details shall apply:

a. Magnitude of test voltage and nature of potential - 350 V ac rms at 60 Hz or 500 V dc.

b. Points of application of test voltage:

Type III - Contact and base plate.

Following this test, connector shall meet the requirements of 3.5.15.

4.7.17 Pin shield retention (types IV and V) (see 3.5.16). Each pin shield shall withstand without separation from its base a force of 4 pounds, minimum. The force shall be applied to each end of each pin shield in a direction normal to a plane passing through the centerline of both keying pins. The point of application of force shall be midway between the first and second electrical contacts on each end and up 1/16 +0, -1/32 inch from the bottom edge of the pin shield. The rate of application of force shall be such that the specified force is obtained in 2 to 5 seconds; the force shall be maintained for a minimum of 10 seconds at 25°C ±5°C (see figure 7). For connectors with more than 40 contacts, a 4 pound minimum force shall also be applied at the center in the same location as shown on figure 7.

4.7.18 Connector fin torque (types IV and V) (see 3.5.17). The connector assembly shall be mounted in a fixture which restrains the movement of the base of the connector assembly (see figure 8). The torque as specified in 3.5.17 shall be applied to the fin in each of two opposing directions. The torque shall be applied at a rate so that the specified torque is obtained in 2 to 10 seconds from the time the torque is first applied to the fin.

4.7.19 Fin cantilever load (type IV) (see 3.5.18). With the connector assembly restrained as specified in 4.7.18, apply the force specified in 3.5.18 at the center of the extractor fin along the centerline between the two extractor holes and perpendicular to the plane of the fin in two directions. The specified force shall be obtained in 2 to 10 seconds from the time the force is first applied.

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4.7.20 Guide rib strength test (types IV and V) (see 3.5.19). The torque specified in 3.5.19 shall be applied in a direction perpendicular to the plane of the ribs and along the full length of the rib using a method similar to that shown on figure 9. The torque shall be applied at a rate so that the specified force is obtained in 2 to 10 seconds from the time the force is first applied to the guide rib.

4.7.21 Temperature-altitude (see 3.5.20). When tested in accordance with MIL-STD-810, method 504.1, category 6, steps 1b, 3, 6, and 13, the connector shall meet the requirements of 3.5.20. When performing step 6, 30 V dc shall be applied between adjacent contacts during the last half hour.

5. PACKAGING

5.1 Packaging shall be in accordance with MIL-C-55330.

6. NOTES

6.1 Intended use. Connectors and component parts covered by this specification are intended for use in ground support, airborne and shipboard electrical and electronic equipment.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Title, number, and date of the applicable specification sheet and the complete part number.

c. Tools (as required).

1. Insertion and withdrawal tools.

2. Crimp tool and positioner.

d. Color (see 3.3.5.3).

e. Marking, levels of preservation and packaging and packing required (see 5.1).

f. Dimensions of base plate and exact number and size of plug connector which the base plate of the type III connector will accommdate (see 3.4).

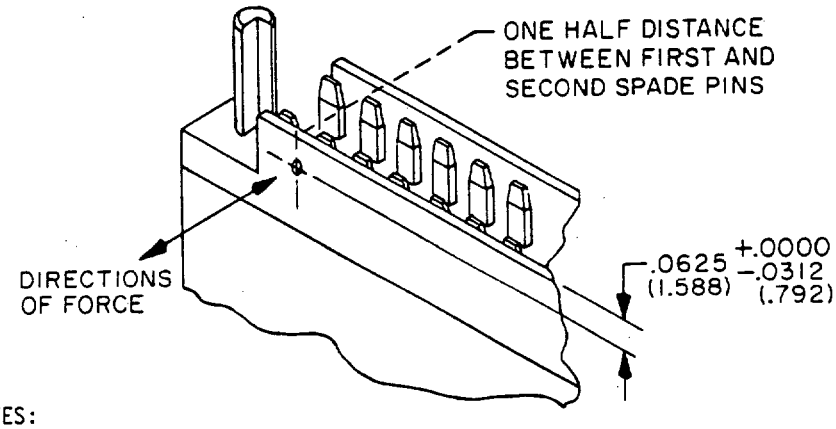
g. OEM drawing must show actual dimensions.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to have the products that they proposed to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360. Information pertaining to qualification of products may be obtained from either the Naval Electronics Systems Command or the Naval Weapons Support Center, Crane, Indiana 47522, agent for administration of the Qualified Products List. Application for qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Provisions Governing Qualification. Copies of "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Current-carrying capacity. The current ratings given in the specification sheets are for single contacts. It is necessary to derate the total current carrying capacity of a connector.

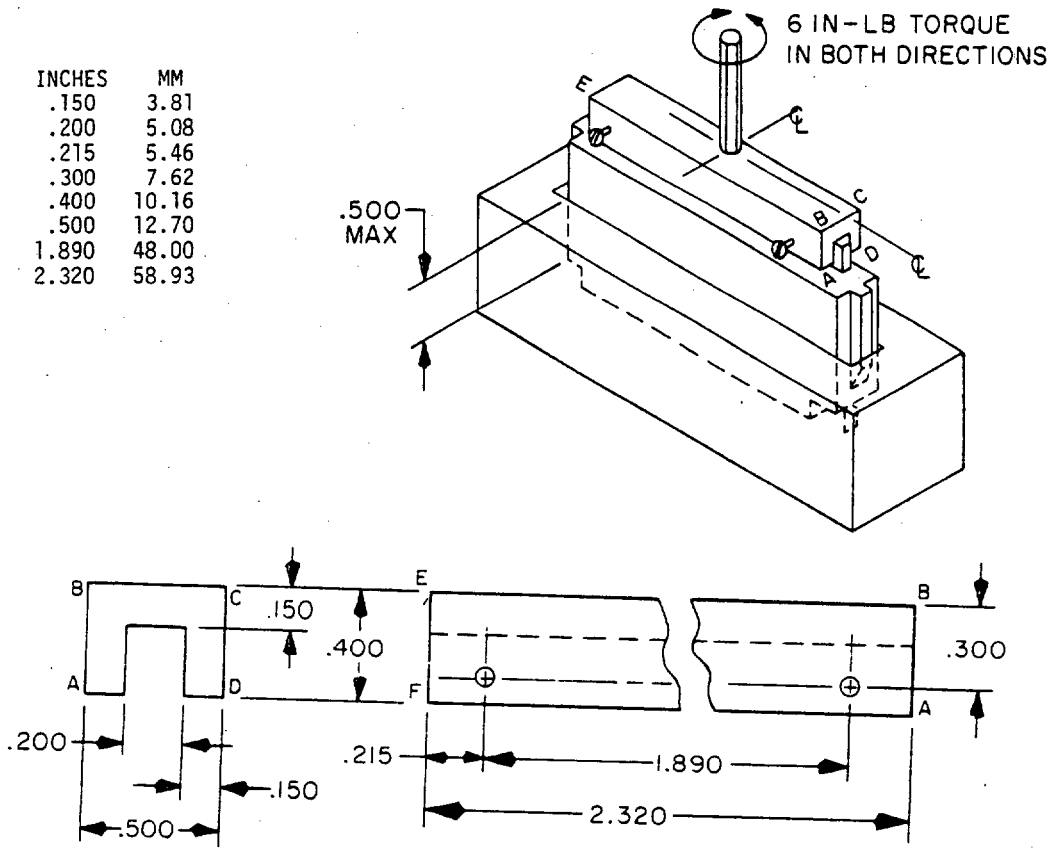
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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Metric equivalents are in parentheses.

FIGURE 7. Pin shield retention test.

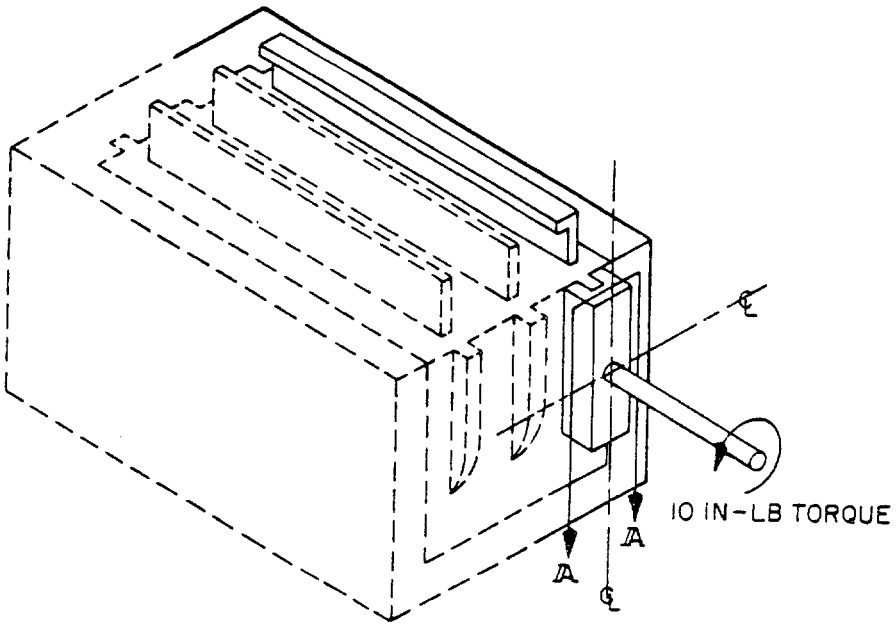
INCHES	MM
.150	3.81
.200	5.08
.215	5.46
.300	7.62
.400	10.16
.500	12.70
1.890	48.00
2.320	58.93

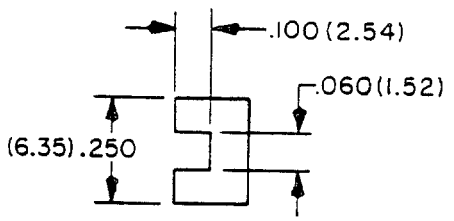


NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 8. Connector assembly fin torque test.

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LENGTH OF SLOT IS 1.286 (32.66)

ENLARGED VIEW A-A

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Millimeters are in parentheses.
4. Unless otherwise specified, tolerances are +.005
   -.000

FIGURE 9. Rib torque test.

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6.5 Shipment of contacts. At the option of the contracting officer, contacts may be mounted on a carrier, or delivered loose in bulk.

6.6 Intermetallic contact. The finishing of metallic areas to be placed in intimate contact by assembly presents a special problem, since intermetallic contact of dissimilar metals results in electrolytic couples which promote corrosion through galvanic action. To provide the required corrosion protection, intermetallic couples are restricted to those permitted by table VII. Table VII shows metals and alloys (or plates) by groups which have common electromotive forces (EMF) within 0.05 volt when coupled with a saturated calomel electrode in sea-water at room ambient temperatures. All members of a group are considered as completely compatible, one with the other. Compatible couples between groups have been specified in table VII based on a potential difference of 0.25 volt maximum. To simplify any arithmetic involved, table VII shows, in addition to EMF against a calomel electrode, a derived anodic index with group 1 (gold, etc.) as 0 and group 18 (magnesium, etc.) as 175. Subtraction of a lower group anodic index gives the EMF difference in hundredths of a volt.

6.6.1 Groups. Table VII sets up 18 primary groups. It may be noted that neither the metallurgical similarity or dissimilarity of metals is the parameter for selection of compatible couples. All members within a group, regardless of metallurgical similarity, are considered inherently nonsusceptible to galvanic action, when coupled with any member within the group; for example, such dissimilar metals as platinum and gold. Similarly, such basically dissimilar alloys as austenitic stainless steel, silver-solder, and low brass (all members of group 5) are inherently nonsusceptible when coupled together.

6.6.2 Compatibility graphs. Permissible couple series are shown in table VII by the graphs at the right. Members of groups connected by lines will form permissible couples. A 0 indicates the most cathode member of each series, a an anodic member, and the arrow indicates the anodic direction.

6.6.3 Selection of compatible couples. Proper selection of metals in the design of equipment will result in fewer intermetallic contact problems. For example, for sheltered exposure, neither silver nor tin require protective finishes. However, since silver has an anodic index of 15 and tin 65, the EMF generated as a couple is 0.50 volt, which is not allowable by table VII. In this case, other metals or plates will be required. It should be noted that, in intermetallic couples, the member with the higher anodic index is anodic to the member with the lower anodic index and will be susceptible to corrosion in the presence of an electrolytic medium. If the surface area of the cathodic part is significantly greated than that of the anodic part, the corrosive attack on the contact area of the anodic part may be greatly intensified. Material selection for intermetallic contact parts, therefore, should establish the smaller part as the cathodic member of the couple, whenever practicable.

6.7 Plating. When base metals intended for intermetallic contact form couples not allowed by table VII, they are to be plated with those metals which will reduce the potential difference to that allowed by table VII.

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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TABLE VII. Compatible couples 1/ (see 6.7)

Group No.	Metallurgical category	EMF (volt)	Anodic index (0, 01 v)	Compatible couples
1	Gold, solid and plated; gold-platinum alloys; wrought platinum (most cathodic)	+0.15	0	
2	Rhodium plated on silver-plated copper	+0.05	10	
3	Silver, solid or plated; high silver alloys	0	15	
4	Nickel, solid or plated; monel metal, high nickel-copper alloys	-0.15	30	
5	Copper, solid or plated; low brasses or bronzes; silver solder; German silver; high copper-nickel alloys; nickel-chromium alloys; austenitic corrosion-resistant steels	-0.20	35	
6	Commercial yellow brasses and bronzes	-0.25	40	
7	High brasses and bronzes; naval brass; Muntz metal	-0.30	45	
8	18 percent chromium type corrosion-resistant steels	-0.35	50	
9	Chromium, plated; tin, plated; 12 percent chromium type corrosion-resistant steels	-0.45	60	
10	Tin-plate; terneplate; tin-lead solder	-0.50	65	
11	Lead, solid or plated; high lead alloys	-0.55	70	
12	Aluminum, wrought alloys of the duralumin type	-0.60	75	
13	Iron, wrought, gray, or malleable; plain carbon and low alloy steels, armco iron	-0.70	85	
14	Aluminum, wrought alloys other than duralumin type; aluminum, case alloys of the silicon type	-0.75	90	
15	Aluminum, cast alloys other than silicon type; cadmium, plated and chromated	-0.80	95	
16	Hot-dip-zinc plate; galvanized steel	-1.05	120	
17	Zinc, wrought; zinc-base die-casting alloys; zinc, plated	-1.10	125	
18	Magnesium and magnesium-base alloys, cast or wrought (most anodic)	-1.60	175	

1/ Compatible couples - potential difference of 0.25 volt maximum between groups.

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Navy - EC
Air Force - 85

Review activities:
Navy - MC
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR, MI
Navy - YD, AS

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5935-3266)

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